

Parth Thakre

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EDUCATION

Purdue University - West Lafayette
Bachelor of Science in Computer Science and Data Science

- Relevant Coursework: Artificial Intelligence, Machine Learning, Algorithms, Object Oriented Programming, Data Structures, Probability, Data Engineering, Computer Architecture, Statistical Theory, Operating Systems

Aug 2022 – Dec 2025
GPA: 3.70

EXPERIENCE

The Data Mine – Purdue University
Software Engineer

- Worked in an agile, sprint-based environment with Microsoft, contributing to iterative development cycles
- Improved development efficiency by 2x by prototyping and testing Minecraft add-ons using vibe coding practices

West Lafayette, IN
Aug 2025 – Dec 2025

Hewlett Packard Enterprise
Data Science Intern

- Built a full-stack team management system (Node.js API + React UI) integrated with LDAP for 5,000+ users, delivering secure RBAC and streamlined CSV-based bulk operations.
- Implemented access control interface with Sequelize ORM and Redux thunks managing 20,000+ product mappings
- Designed REST API with server-side caching and pagination for large-scale enterprise datasets

Spring, TX
May 2025 – Aug 2025

Purdue Vertically Integrated Projects (VIP)
Machine Learning Engineer

- Developed coefficient of variation-based algorithm reducing AP-10K data by 82% while improving Average Precision by 5%, accelerating RTMPose training from 2.5-3 hours to 25-30 minutes
- Implemented keypoint labeling pipeline with Label Studio to enhance pose detection accuracy
- Paper accepted into WACV 2026 conference for publishing

West Lafayette, IN
Aug 2024 – Aug 2025

Purdue University
Data Science Researcher

- Analyzed 35+ years of IMPROVE aerosol data from Midwest sites using plots for time series, regressions, Mann-Whitney U and Mann-Kendall tests to identify spikes in aerosol compositions due to wildfires
- Developed 2000+ line Python pipeline to quantify OC:EC ratio trends utilizing statistical testing

West Lafayette, IN
Aug 2024 – Aug 2025

SiTime
Software Development Intern

- Reduced wafer test analysis from 2.5+ hours to 3-8 minutes by implementing Rust-backed STDF parsing with optimized pandas transformations, accelerating failure detection by 90%
- Engineered PyQt6 dashboard with vectorized operations processing multi-GB files across 3M+ records, enabling real-time manufacturing anomaly identification

Santa Clara, CA
Jul 2024 – Aug 2024

PROJECTS

AI Risk Assessment | *React, Next.js, FastAPI, Python, SQLAlchemy, LLM*

- Designed and implemented a 3-stage LLM security pipeline with FastAPI asyncio across 10+ endpoints
- Utilized async system to retrieve prompts from the database, send to LLM, and automatically grade using LLMs
- Built a full-stack platform using Next.js + Tailwind CSS frontend with FastAPI + SQLAlchemy backend APIs

Bull-It | *React, FastAPI, LLM, Python, XGBoost*

- Engineered ML stock predictor achieving 84% accuracy using XGBoost, LSTM, RSI, MACD, Bollinger Bands
- Trained and validated models on 20+ years of market data using robust train, validation, and test splits
- Built full-stack application with FastAPI backend, React frontend, and AWS S3 for secure storage and retrieval

TECHNICAL SKILLS

Languages: Python, Java, C++, R, SQL, JavaScript (Node.js, React), SAS, HTML/CSS
Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, XGBoost, Hugging Face, VADER, RTMPose
Data Analysis & Visualization: pandas, NumPy, Matplotlib, Seaborn, ggplot2, Tableau, PyQt6
Tools & Platforms: FastAPI, REST APIs, AWS S3, SSMS, Git/GitHub, Docker, Linux, SAP, LabelStudio